FIT2179 Data Visualisation

Course Notes

# FLOW MAPS

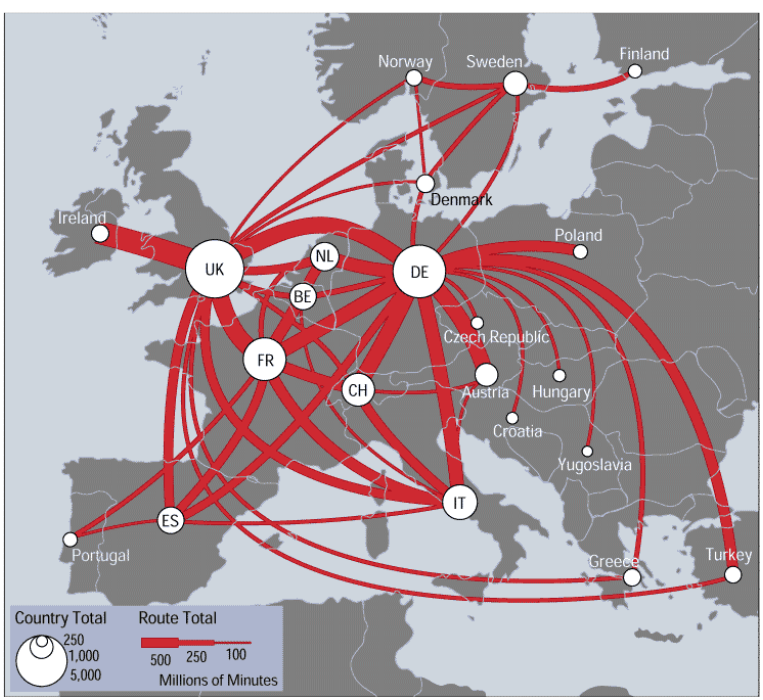
13 September 2026

# FLOW MAPS

Flow maps show movement. There are three main types: origin–destination flow maps, maps that show the geometry of paths, and maps that show vector fields.

Origin–destination flow maps are spatial Sankey diagrams of networks; the geometry of links is not relevant.

The second type does show real geometry, as in a map of flight paths. The third type visualises vector fields, for example air movement drawn with tightly packed arrows.



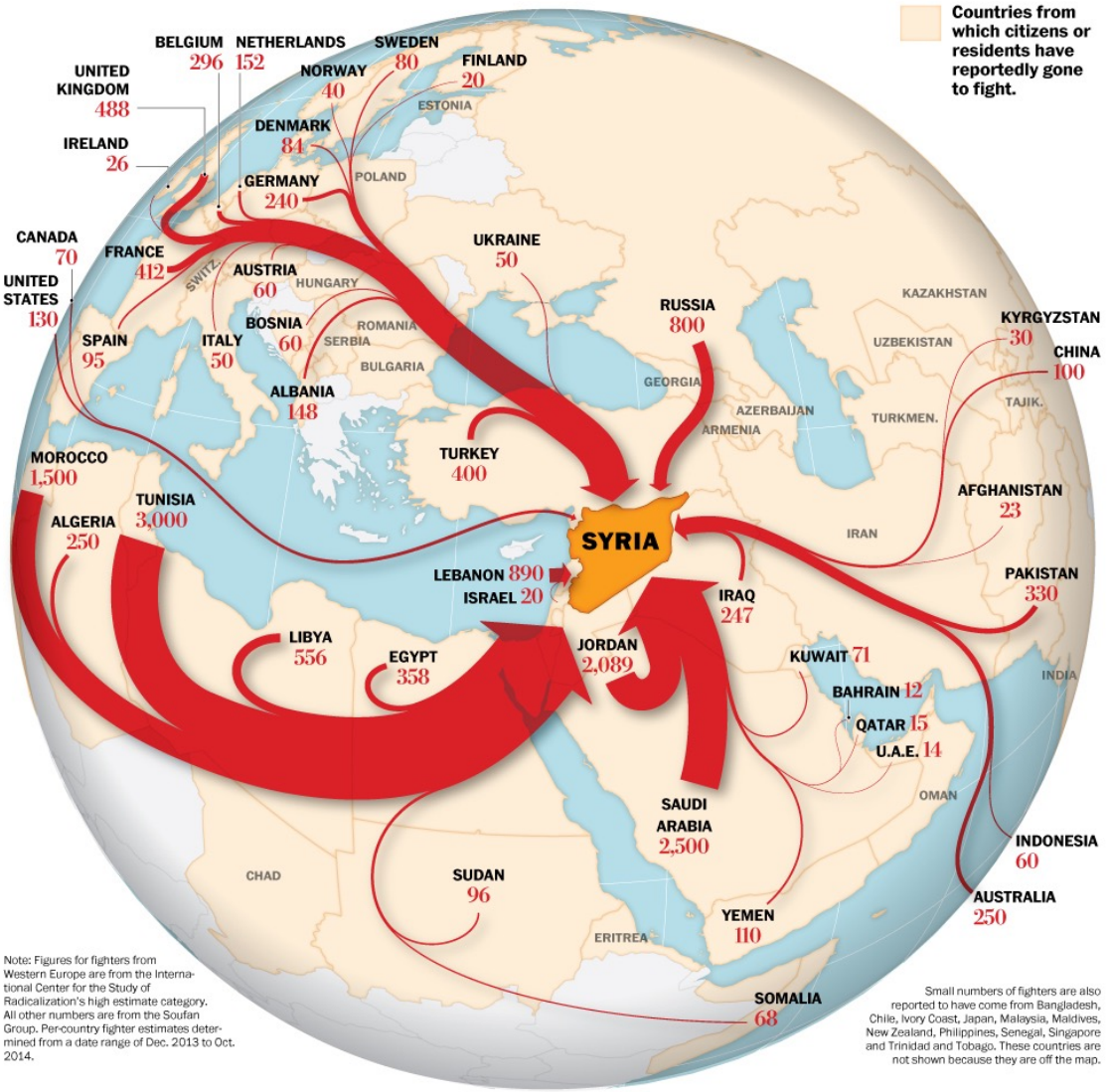

## ORIGIN–DESTINATION FLOW MAPS

Origin–destination flow maps show flows in a network as straight or curved lines. They do not aim to show the exact geometry between origin and destination locations; the focus is on the quantity or quality of what flows through the network.

Line width can be varied to indicate quantitative attributes. A line does usually not represent a single item, but the sum of items flowing between two nodes.

The map shows telecommunication traffic, measured in minutes, between European countries. The exact position and geometry of the telecommunication infrastructure is irrelevant. Only the amount of communication matters, so the author is free to arrange the flow lines in whatever way optimises readability.

Up to about 50, or perhaps 100, flows can be shown on a single origin-destination flow map. Beyond that, overlapping flows tend to produce cluttered maps that are difficult to read.




Converging or diverging flows can be shown to indicate regional totals. An excellent example shows the origins and numbers of foreign fighters in the Syrian war in 2014. Merging makes it easy to compare the number of fighters from Europe with the number from North Africa, for instance. Constructing merged flow lines of this kind is not simple, and might require vector editing software, such as Adobe Illustrator.

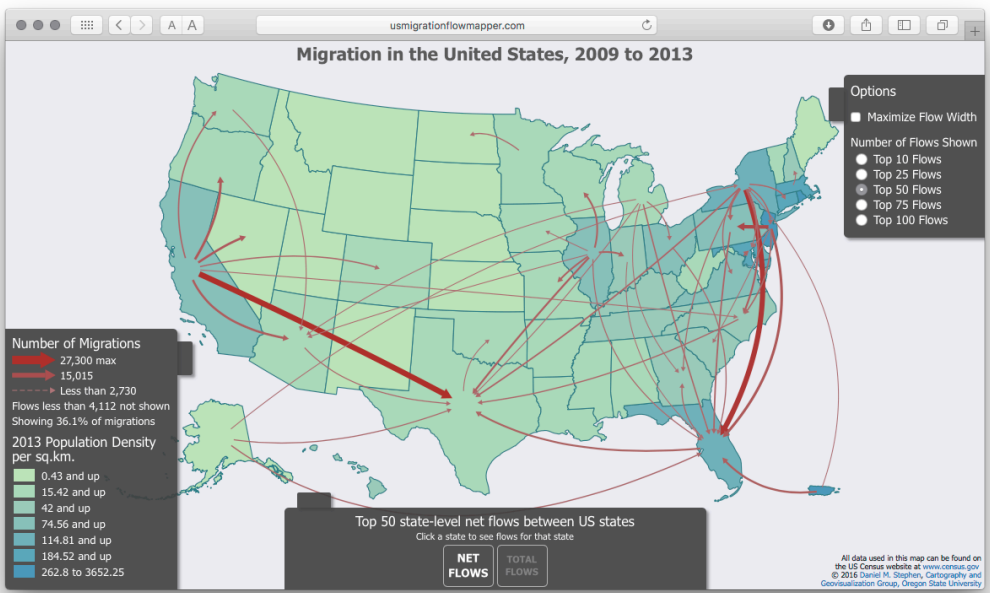
Map by Washington Post October 2014

What can be automated is shown by a map that uses a force-directed algorithm to bend flow lines. The approach resembles algorithms for node-link diagrams, except that node positions are fixed. Flows push each other apart, minimising overlaps and improving readability. The map is available at usmigrationflowmapper.com.

## FLOW MAPS SHOWING PATH GEOMETRY

The second type shows individual moving items such s animals tracked by GPS, or aircraft, as in the example here. Each line represents one item. Bundling algorithms are useful for this type of map.





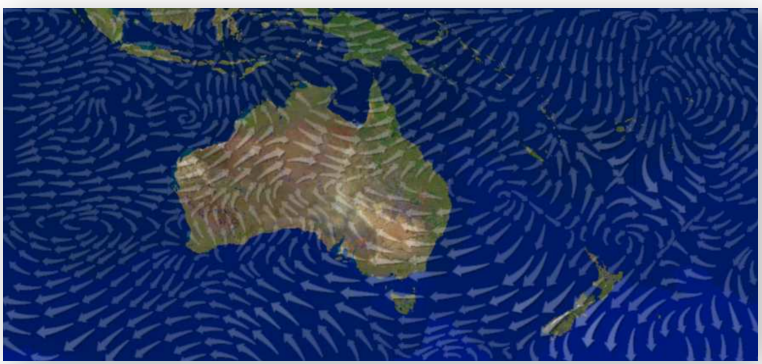

usmigrationflowmapper.com

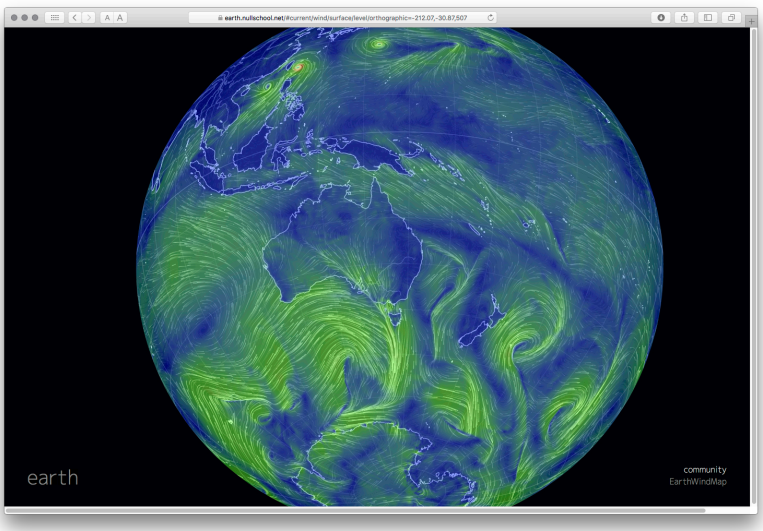
## VECTOR FIELD FLOW MAPS

The third type shows vector fields: spatial fields containing direction information and possibly further attributes. Fields can be two- or three-dimensional.

The algorithms behind these visualisations extract flow directions from the vector field, tracing a line by following the directions stored in the field's cells. The extracted lines are then visualised either by placing a streamline glyph along the line, or by smearing the pixels of a raster image along its direction. Both techniques can produce aesthetic and effective results. The ocean current map uses an algorithm that selects streamlines so that they are evenly spaced.

The pixel-smearing method appears in an interactive globe at earth.nullschool.net, which is enjoyable to explore. There, the flow visualisation is combined with other scalar fields, such as air temperature, encoded with colour.

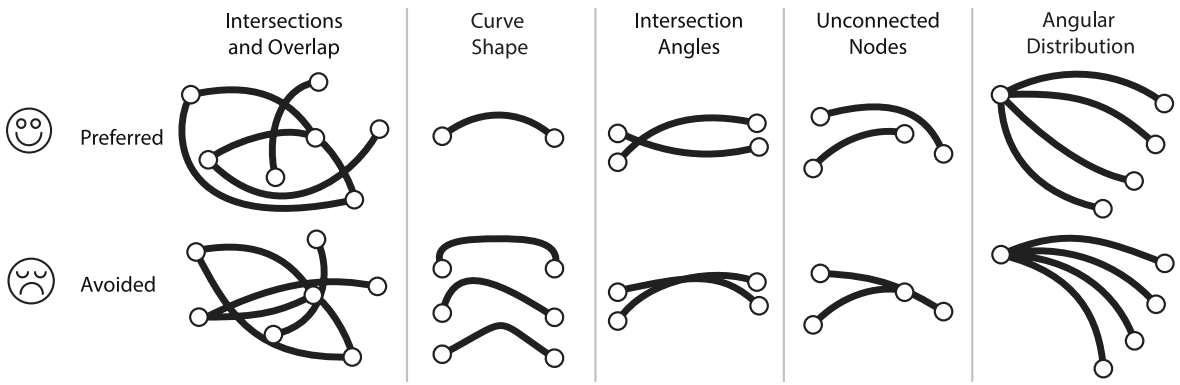





earth.nullschool.net

## DESIGN PRINCIPLES

More specific design guidelines for placing flow lines on origin–destination flow maps follow below.




Source: Jenny et al. (2018). Design principles for origin-destination flow maps. Cartography and Geographic Information Science.